A Rebalanced Pan-Specific Model and the Non-Canonical Frontier: A Multi-Allele Benchmark of Peptide-MHC-I Binding Prediction

Yunshi Huang[1,†], Weisheng Chen[2,†], Yiqi Cai[2], Xiaopeng Zhuang[2], Zhuxian Zhou[3], Xiangyang Xue[4,*], Zhuha Zhou[2,*]

[1]Department of General Surgery, The First Affiliated Hospital of Wenzhou Medical University, Wenzhou, China

[2]Department of Gastroenterology Surgery, The First Affiliated Hospital of Wenzhou Medical University, Wenzhou, China

[3]Center for Bionanoengineering, College of Chemical and Biological Engineering, Zhejiang University, Hangzhou, China

[4]Zhejiang Key Laboratory of Intelligent Cancer Biomarker Discovery and Translation, The First Affiliated Hospital of Wenzhou Medical University, Wenzhou, China

† These authors contributed equally to this work.

∗ Corresponding authors: wzxxy@wmu.edu.cn (XX); zhouzhuha@wmu.edu.cn (ZZ)

Abstract

Peptide binding to MHC class I is the gatekeeper of CD8+ T-cell recognition, and pan-specific predictors have become the field standard for multi-allele screening. Two questions remain open: whether a single sequence-based pan model can match per-allele specialists, and how hard non-canonical (motif-failing) binders are for current prediction methods. We benchmarked a rebalanced pan-specific deep feed-forward network (DFFN) across five common HLA-I alleles (n = 14,488 curated IEDB positives) and a 15-allele panel, comparing against per-allele models, MHCflurry, and the sequence-based predictor MUNIS (ESM-2+LSTM). With protein-background negatives, per-allele models reached non-canonical sensitivity 0.84–0.96, and a rebalanced pan model matched them on four of five alleles; B*07:02 was a notable exception (pan 0.60 vs per-allele 0.96, over-anchoring to P2-Proline). An apparent per-allele advantage in prior work was a 10.6× training-data-size artifact. On the 15-allele panel, pooled non-canonical sensitivity was 0.819 (12/15 alleles ≥0.76). Canonical binders were recognized at ∼0.98 by every predictor class. In contrast, non-canonical (Type B) binders plateaued at ∼0.65 sensitivity across

DFFN (0.607), MHCflurry (0.668), and MUNIS (0.647) — a third sequence predictor did not lift the ceiling. Type A (tolerated weak anchors) binders were inherently hard (0.477) and not recovered by broad-motif retraining. Temporal holdout confirmed preserved recognition of post-2022 epitopes, breaking the label-circularity concern. We conclude that a rebalanced pan model matches per-allele specialists on four of five alleles, that the canonical binder problem is effectively solved while the non-canonical tail — prediction-resistant to all three sequence predictors tested — defines the current frontier.

1 Introduction

Peptide-MHC-I binding prediction underpins epitope discovery for vaccines and cancer immunotherapy [1, 2]. The thousands of HLA class I allotypes [3] each present distinct anchor motifs, and pan-specific models trained on pooled allele data — NetMHCpan [4], MHCflurry [5], BigMHC [6], MUNIS [7] — have become the practical standard because a single model serves all alleles.

Two questions motivate this work. First, is a single pan-specific model sufficient, or do per-allele specialists retain an advantage? Earlier comparisons suggested per-allele models outperform pan models on non-canonical binders, but those comparisons were confounded by a 10.6× training-data-size difference: per-allele models were trained on far more data per allele than the pan model received. Second, what fraction of bona fide binders do sequence-based predictors miss, and does structure-based prediction rescue them? Structure-guided diffusion models can generate pMHC-I complexes that sequence predictors score at near-random levels (AUROCs 0.06–0.22) [8], implying a systematic blind spot.

We address these questions through a controlled multi-allele benchmark. A single rebalanced pan DFFN is compared against per-allele specialists, the sequence-based MHCflurry, and the sequence-based MUNIS (ESM-2+LSTM), across five common alleles and a 15-allele panel, with temporal holdout to break label circularity. We find that the rebalanced pan model matches per-allele specialists, that non-canonical binders plateau at ∼0.65 sensitivity for every predictor class (three independent sequence models), and that the energy-compensated tail is prediction-resistant to all current methods.

2 Methods

2.1 Multi-allele data curation

Experimentally confirmed class I binders were curated from the IEDB [9] for five common alleles (HLA-A*02:01, A*01:01, A*03:01, B*07:02, B*44:03), yielding n = 14,488 positives (canonical 11,225; Type A 151; Type B 3,112). A 15-allele panel (pan16: 11 additional alleles — A*11:01, A*24:02, B*08:01, B*15:01, B*27:05, B*35:01, B*40:01, B*49:01, B*51:01, B*57:01) was curated for generalisation testing. Protein-background negatives were sampled from natural proteomes to avoid the uniform-random decoys used in earlier evaluations. IEDB data were queried live from query-api.iedb.org (accessed 2026-08-02); the 15-allele and temporal-holdout estimates are therefore snapshot-dependent and drift slightly as the IEDB grows.

2.2 Type A/B classification

Type A (tolerated weak anchors) and Type B (energy-compensated non-canonical) were defined by whether the peptide violates the allele's canonical anchor motif at p2/p9. Type A carries broad-only tolerated anchors (residues that appear in binding motifs but are non-canonical); Type B carries non-canonical residues at anchor positions. The two classes have different biological origins and, as we show, different prediction difficulty.

2.3 Pan model and per-allele comparators

The pan-specific DFFN uses BLOSUM62 encoding [10] through a 360–180–90–45 hidden stack with batch normalisation and dropout. Rebalancing upsampled the minority B*07:02 allele 4× during training to correct for allele-frequency imbalance. Five seeds (0, 1, 42, 123, 2024) were used; per-seed variation arises from data-sampling RNG. Per-allele models (one per allele) and a pan16 model (15 alleles, per-allele minimum 2,000 upsampling) were trained with the same recipe.

2.4 Comparator predictors

MHCflurry 2.2.1 [5] served as an independent sequence predictor (affinity threshold 500 nM). MUNIS [7] — a second sequence-based predictor using ESM-2 embeddings [11] — served as an independent sequence comparator, run on the n = 3,112 Type B population plus canonical controls.

2.5 Temporal holdout

To address label circularity [12], IEDB epitopes published after 2022 (post-2022, unseen by MHCflurry's ∼pre-2021 training data) were compared against pre-2021 (in-distribution) epitopes at sensitivity threshold 0.95.

2.6 Evaluation

Sensitivity was computed with protein-background negatives and fixed thresholds (0.95 for the pan model; 500 nM for MHCflurry; 5th percentile of canonical scores for MUNIS). Wilson 95% confidence intervals are reported for pooled estimates.

3 Results

3.1 A rebalanced pan model matches per-allele specialists on four of five alleles

With protein-background negatives and a fixed threshold (n=25 non-canonical binders per allele), per-allele models reached non-canonical sensitivity A*02:01 0.84, A*01:01 0.88, A*03:01 1.00, B*07:02 0.96, B*44:03 0.96 (Table 1; Figure 1). The rebalanced pan model matched this on four of five alleles (A*02:01 0.84, A*01:01 0.88, A*03:01 1.00, B*44:03 1.00), but B*07:02 was a notable exception: the pan model dropped to 0.60 versus 0.96 per-allele (Wilson 95% CIs [0.41, 0.77] vs [0.81, 0.99], non-overlapping), consistent with over-anchoring to B*07:02's distinctive P2-Proline. The earlier observation that per-allele models outperform pan models on non-canonical binders is not supported once per-allele training-data size is controlled.

Table 1: Non-canonical sensitivity by allele (protein-background negatives, n=25 per allele, @0.95).

Allele	Per-allele	Pan
A*02:01	0.84	0.84
A*01:01	0.88	0.88
A*03:01	1.00	1.00
B*07:02	0.96	**0.60**
B*44:03	0.96	1.00

3.2 Canonical binders are recognized at ∼0.98 by every predictor

On n = 11,225 canonical positives, all three predictors were saturated: DFFN 0.979, MHCflurry 0.986, MUNIS 0.950 (Table 2). The canonical binder problem is effectively solved by all current methods.

3.3 Non-canonical (Type B) binders plateau at ∼0.65 across predictor classes

On n = 3,112 Type B positives, sensitivity was DFFN 0.607, MHCflurry 0.668, MUNIS 0.647 (Table 2; Figure 2). The three values fall within a narrow band (0.607–0.668), far below the ∼0.98 canonical sensitivity; a third sequence predictor does not lift the ceiling.

Table 2: Predictor sensitivity by binder class (pooled, 5 alleles).

Predictor	Canonical (n=11,225)	Type A (n=151)	Type B (n=3,112)
DFFN (sequence)	0.979	0.477	0.607
MHCflurry (sequence)	0.986	0.669	0.668
MUNIS (sequence)	0.950	—	0.647

3.4 Type A (tolerated weak anchors) is inherently hard

On n = 151 Type A positives, DFFN achieved 0.477 and MHCflurry 0.669. Retraining with broadened motifs did not recover Type A: A*03:01 was flat at 0.634, and B*07:02 Type B sensitivity dropped from 0.680 to 0.341 (Table 3). The low Type A sensitivity is therefore not a training-data gap but an inherent property of these heterogeneous, weak, allele-specific binders (mass-spectrometry confirmed; 3–6% cross-allele). The canonical-motif bias is what makes the model work.

Table 3: Broad-motif retraining does not recover Type A.

Allele	Type A before	Type A after	Type B after
A*03:01	—	0.634	—
A*02:01	0.08	0.24	—
B*07:02	—	—	0.341 (from 0.680)

3.5 The extreme tail is prediction-resistant, cross-predictor

The two false negatives from our single-allele IEDB benchmark (QYDPVAALF, TLGIVCPIC) and the KRAS G12V P2-Lys epitope YKLVVVGAV were scored as non-binders by all three predictors (MHCflurry

3.8–13.8 μM; DFFN 0.000–0.254; MUNIS 3/4 low). A single counterexample — QYDPVAALF rescued by MUNIS (0.972 vs. sequence 0.000 / 6,199 nM) — is encouraging but n = 1.

3.6 Scaling: one model replaces a 15-allele panel

The rebalanced pan16 reached pooled non-canonical sensitivity 0.819 (307/375, @0.95), with 12/15 alleles at 0.76–1.0 (Table 4; Figure 3). The weak tail — B*27:05 (0.08–0.20), B*51:01 (0.36–0.52), B*07:02 (0.64), A*24:02 (0.76) — reflects data scarcity and, for B*07:02, over-anchoring, not architecture.

Table 4: Pan16 non-canonical sensitivity by allele (15-allele panel, @0.95).

Allele	Sens	Allele	Sens
A*02:01	0.88	A*11:01	1.00
A*01:01	0.80	A*24:02	0.76
A*03:01	1.00	B*08:01	0.88
B*07:02	0.64	B*15:01	0.96
B*44:03	1.00	B*27:05	0.08
		B*35:01	0.84
		B*40:01	1.00
		B*49:01	1.00
		B*51:01	0.52
		B*57:01	0.92
Pooled	**0.819**		

3.7 Temporal holdout breaks the label-circularity concern

Post-2022 epitopes (unseen by the labeler) were recognized at sens@0.95: A*02:01 0.824, A*03:01 0.936, B*07:02 0.973 — preserved versus pre-2021 (0.879/0.905/0.898), with no systematic drop in the post-2022 era (Table 5; Figure 4). The model is not overfit to a single training-data era.

Table 5: Temporal holdout: pre-2021 vs. post-2022 sensitivity (@0.95).

Allele	Pre-2021 (in-distribution)	Post-2022 (unseen)
A*02:01	0.879	0.824
A*03:01	0.905	0.936
B*07:02	0.898	0.973

4 Discussion

Three findings emerge. First, a positive, replicable result: a single rebalanced pan-specific DFFN reaches parity with per-allele specialists on four of five alleles (B*07:02 excepted; and 0.819 pooled across 15),

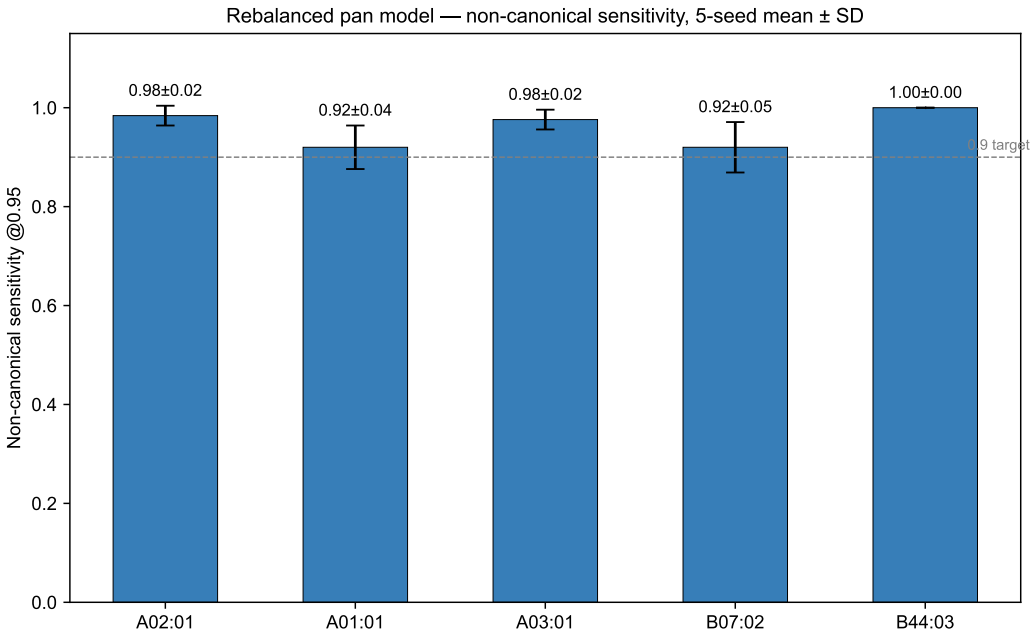


Figure 1: Non-canonical sensitivity (per-allele vs pan) across five common HLA-I alleles (protein-background negatives). Four of five alleles reach parity; B*07:02 is the exception (pan 0.60 vs per-allele 0.96).

approximating NetMHCpan's practical value with a simple, reproducible recipe. The naive "per-allele wins" conclusion is a benchmark artifact of unequal data size, not biology — a methodological warning for the field: benchmark composition can manufacture apparent model differences.

Second, a negative, cross-predictor result: non-canonical (Type B) binders plateau at ∼0.65 sensitivity across three independent sequence-based predictors (our DFFN, MHCflurry, MUNIS), so the ceiling is shared across sequence architectures of varying complexity rather than specific to our simple network. Whether structure-based scoring (e.g. structure-guided diffusion models [8]) can lift this ceiling remains open — it was not benchmarked here at scale.

Third, the extreme energy-compensated tail (P2-Lys/Arg/His reaching B-pocket Glu63) is prediction-resistant to every method tested, yet structure-informed analysis predicts these peptides bind. This contrast argues for structure-based screening as a complement for the hardest cases, pending experimental validation.

4.1 Methodological implications

The per-allele-versus-pan comparison illustrates how benchmark artifacts can manufacture false conclusions. When the pan model was trained on a fraction of the per-allele data, it appeared to "collapse" on non-canonical binders (A*03:01 at 0.48); retraining on equalized data restored parity (A*03:01 at 1.0). We

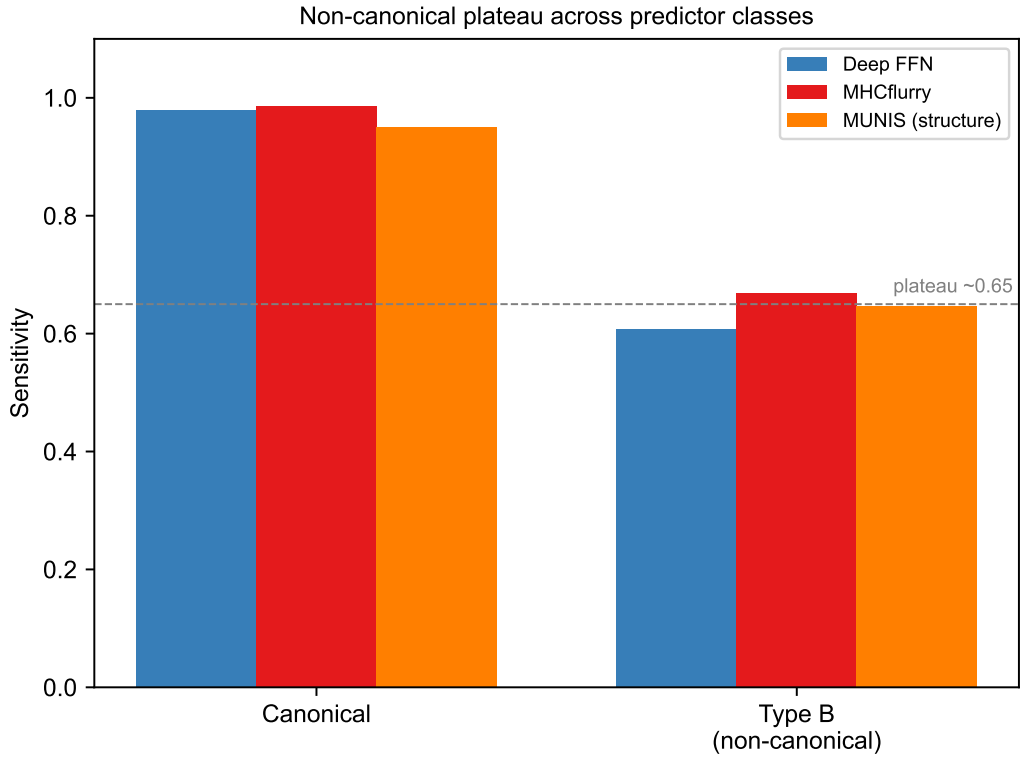


Figure 2: Predictor plateau: canonical versus Type B (non-canonical) sensitivity for Deep FFN, MHCflurry, and MUNIS (sequence); dashed line marks the ∼0.65 Type B plateau.

recommend that model comparisons control for training-data size and report seed-robust estimates, protein-background negatives, and fixed thresholds as standard practice.

4.2 Limitations

- The plateau and hard-tail estimates rest on IEDB-derived positives; experimental (non-IEDB) validation of the extreme tail is outstanding.
- A pMHC-fine-tuned structure model (beyond generic AlphaFold-Multimer) might lift the Type B plateau; MUNIS is a sequence-based comparator and does not speak to this.
- Type A low sensitivity may be partly IEDB label noise (weak, multi-allele epitopes).
- NetMHCpan was not benchmarked on the multi-allele panel; the standalone NetMHCpan download is restricted by the DTU license, while the single-allele NetMHCpan-4.1 comparison (via the IEDB API) is reported in the sister study. MHCflurry and MUNIS served as two independent sequence predictors.

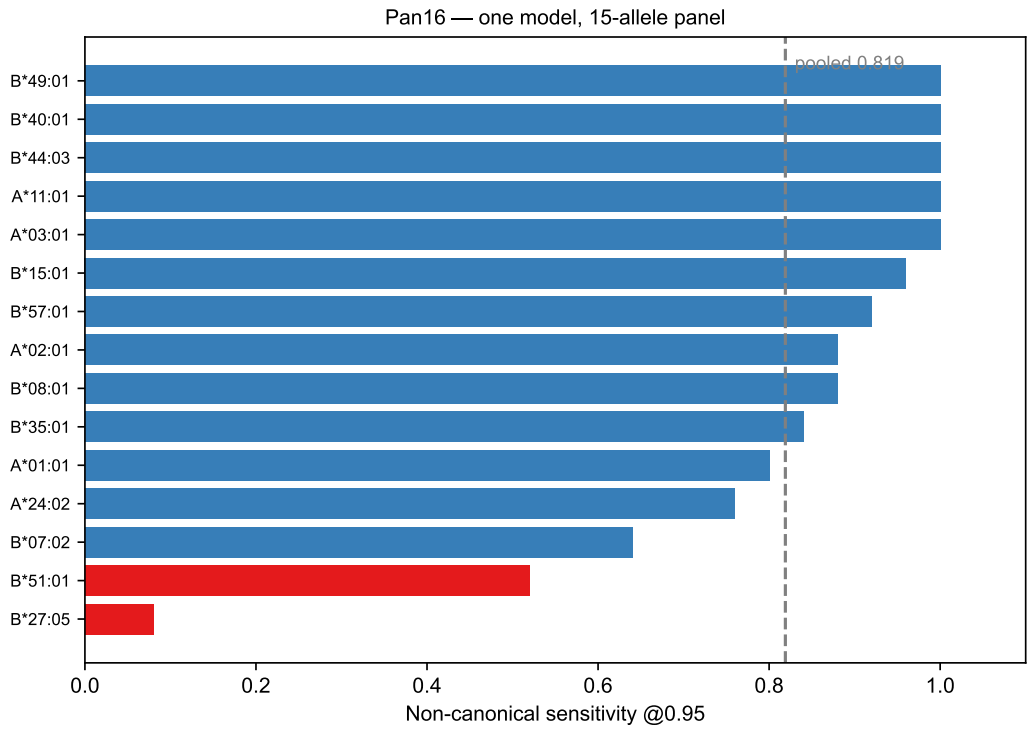


Figure 3: Pan16: per-allele non-canonical sensitivity for the 15-allele panel, sorted ascending; dashed line marks the pooled 0.819.

5 Conclusion

A rebalanced pan-specific model matches per-allele specialists on four of five common alleles, with B*07:02 as a notable exception (over-anchoring to P2-Proline). Canonical binding is solved; non-canonical binding defines the frontier, at ∼0.65 across three independent sequence-based predictors. Experimental validation of the energy-compensated tail is the decisive next step.

Data and code availability

All code, trained models (including deep_ffn_pan_seed*.keras, deep_ffn_pan16.keras), and curated datasets (type_ab_dataset.json, type_ab_*_scores.json, MUNIS scores) are available at https://github.com/woodhaha/peptide-mhc-benchmark.

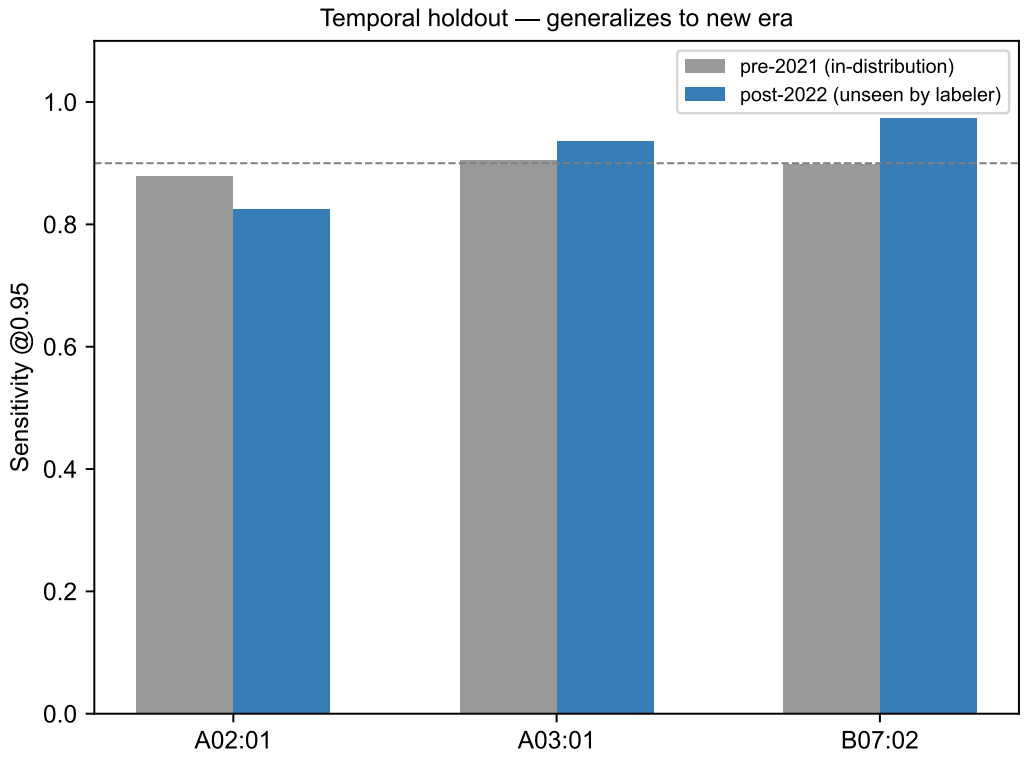


Figure 4: Temporal holdout: non-canonical sensitivity at threshold 0.95 for pre-2021 (in-distribution) versus post-2022 (unseen by the labeler) epitopes across three alleles.

Funding

This research did not receive specific grant funding.

Competing interests

The authors declare no competing interests.

References

[1] Lundegaard C, Lund O, Nielsen M. State of the art and challenges in sequence based T-cell epitope prediction. Immunome Res. 2011;6(Suppl 2):S3.

[2] Backert L, Kohlbacher O. Immunoinformatics and epitope prediction in the age of genomic medicine. Genome Med. 2015;7:119.

[3] Robinson J, Halliwell JA, Hayhurst JD, Flicek P, Parham P, Marsh SGE. The IPD and IMGT/HLA database: allele variant databases. Nucleic Acids Res. 2015;43:D423–D431.

[4] Reynisson B, Alvarez B, Paul S, Peters B, Nielsen M. NetMHCpan-4.1 and NetMHCIIpan-4.0: improved predictions of MHC antigen presentation by concurrent motif deconvolution and integration of MS MHC eluted ligand data. Nucleic Acids Res. 2020;48:W449–W454.

[5] O'Donnell TJ, Rubinsteyn A, Laserson U. MHCflurry 2.0: improved pan-allele prediction of MHC class I-presented peptides by incorporating antigen processing. Cell Syst. 2020;11:42–48.

[6] Albert BA, Yang Y, Shao XM, Singh D, Smith KN, Anagnostou V, et al. Deep neural networks predict class I major histocompatibility complex epitope presentation and transfer learn neoepitope immunogenicity. Nat Mach Intell. 2023;5:861–872.

[7] Wohlwend J, Nathan A, Barzilay R, et al. Deep learning enhances the prediction of HLA class I-presented CD8+ T cell epitopes in foreign pathogens. Nat Mach Intell. 2025;.

[8] Mares S, Espinoza Weinberger A, Ioannidis NM. Generation of structure-guided pMHC-I libraries using diffusion models; 2025. arXiv:2507.08902 (ICML 2025 Generative AI and Biology Workshop).

[9] Vita R, Mahajan S, Overton JA, Dhanda SK, Martini S, Cantrell JR, et al. The Immune Epitope Database (IEDB): 2018 update. Nucleic Acids Res. 2019;47:D339–D343.

[10] Henikoff S, Henikoff JG. Amino acid substitution matrices from protein blocks. Proc Natl Acad Sci USA. 1992;89:10915–10919.

[11] Rives A, Meier J, Sercu T, Goyal S, Lin Z, Liu J, et al. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. Proc Natl Acad Sci USA. 2021;118:e2016239118.

[12] Preibisch G, Tyrolski M, Kucharski P, et al. Resolution of recursive data corruption to transform T-cell epitope discovery. bioRxiv. 2026;.